

N100 Financial Intelligence Platform

Final Project Report

Submitted By:

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Project Duration:

Day 1 – Day 28

Technologies Used:

Python, Pandas, NumPy, SQLite, SQL Alchemy, Matplotlib, Seaborn, Plotly, Power BI, GitHub.

Table of Contents

1. Introduction
2. Platform Overview
3. System Architecture
4. Installation and Setup
5. Running the Application
6. Dashboard Navigation Guide
7. Screener Module Usage
8. Company Profile Analysis
9. Peer Comparison Module
10. Valuation & Market Data Analysis
11. PDF Tearsheet Generation
12. API Usage Guide
13. Data Quality and Analytics Engine
14. Troubleshooting Guide
15. Developer Reference
16. Conclusion

1. Introduction

1.1 Purpose of the Platform

The N100 Financial Intelligence Platform is an end-to-end financial analytics solution designed to provide investors, analysts, and researchers with a comprehensive view of Nifty 100 companies.

The platform combines:

- Automated ETL pipelines
- Financial KPI calculations
- Company screening
- Peer comparison
- Sector analysis
- Portfolio statistics

- Clustering analysis
- API services
- Interactive dashboards

The objective is to transform raw financial data into meaningful investment insights.

1.2 Key Features

The platform provides:

Financial Analysis

- Profitability analysis
- Growth analysis
- Balance sheet evaluation
- Cash flow assessment
- Capital allocation analysis

Screening Engine

Users can identify companies based on:

- ROE
- Debt-to-equity ratio
- Revenue growth
- PAT growth
- Free cash flow
- Valuation metrics

Peer Analysis

The platform compares companies against their industry peers using:

- Percentile ranking
- Financial ratios
- Growth metrics
- Quality scores

Dashboard

Interactive Streamlit dashboard provides:

- Company overview
- Sector analysis
- Screener results
- Peer comparison
- Trend analysis
- Reports

2. Platform Overview

The platform processes financial information from multiple sources and converts it into analyst-ready insights.

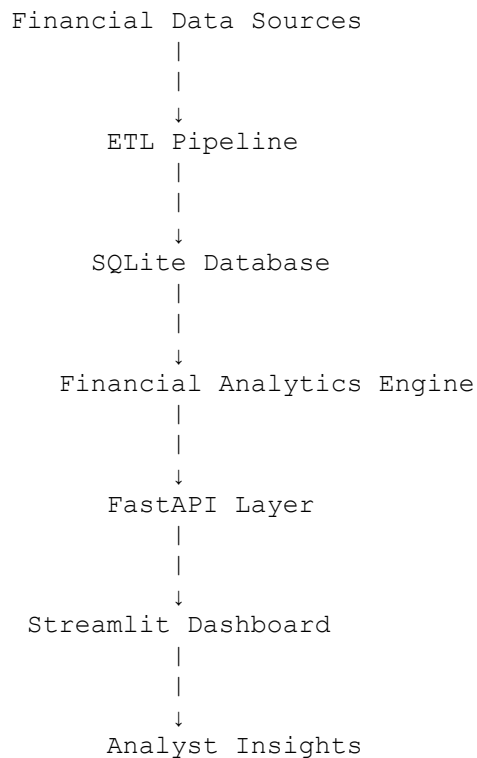
Main Components

Component	Purpose
ETL Pipeline	Data ingestion and cleaning

Component	Purpose
SQLite Database	Data storage
KPI Engine	Financial calculations
Screener Engine	Company filtering
Analytics Engine	Clustering and statistics
FastAPI	Backend services
Streamlit	User interface
Testing Framework	Quality validation

3. System Architecture

The complete workflow:



4. Installation and Setup

4.1 Requirements

Required software:

Python 3.13+

SQLite

Git

Streamlit

FastAPI

Uvicorn

4.2 Clone Project

Navigate to project location:
N100_Financial_Intelligence_Platform

4.3 Create Virtual Environment

Command:

```
python -m venv venv
```

Activate:

```
venv\Scripts\activate
```

4.4 Install Dependencies

Run:

```
pip install -r requirements.txt
```

Scheme Name

Fund House

Category

Sub-Category • Risk Grade

5. Running the Application

5.1 Run ETL Pipeline

The ETL pipeline loads financial data into SQLite.

Command:

```
python src/etl/loader.py
```

Output:

```
db/nifty100.db
```

5.2 Run Analytics Modules

Example:

Cluster generation:

```
python src/analytics/clustering.py
```

Generated outputs:

```
output/cluster_labels.csv
```

```
reports/elbow_plot.png
```

6. Streamlit Dashboard Guide

6.1 Start Dashboard

Command:

```
streamlit run src/dashboard/app.py
```

Open browser:

```
http://localhost:8501
```

Dashboard Screens

The dashboard contains eight major screens.

Screen 1 — Home Dashboard

Purpose

Provides overall market summary.

Displays:

- Total companies covered
 - Sector distribution
 - Portfolio statistics
 - Market overview
-

Screen 2 — Company Profile

Purpose

Detailed analysis of individual companies.

Users can search by:

- Company name
- Ticker symbol

Example:

TCS

INFY

HDFCBANK

Displayed information:

- Company details
 - Sector information
 - Latest KPIs
 - Financial trends
-

Screen 3 — Screener Module

Purpose

Find companies matching investment criteria.

Available filters:

- ROE
- Debt-to-equity
- Revenue CAGR
- PAT CAGR
- Free cash flow

- Sector

Example Screener

Quality Compounder:

ROE > 15%

Debt/Equity < 1

Revenue CAGR > 10%

FCF Positive

The screener returns:

- Company name
- Sector
- KPI values
- Ranking score

Screen 4 — Peer Comparison

Purpose

Compare companies against similar businesses.

Metrics:

- ROE percentile
- Growth percentile
- Margin percentile
- Leverage percentile

The radar chart provides visual comparison between:

- Selected company
- Peer average
- Benchmark company

Screen 5 — Trends Analysis

Shows historical performance:

Available trends:

- Revenue growth
- Profit growth
- EPS growth
- Margin trends
- Cash flow trends

Period:

2019 - 2024

Screen 6 — Sector Analysis

Provides sector-level insights.

Displays:

- Number of companies
- Median ROE
- Median valuation
- Debt levels

Covered sectors:

- IT
- Banking
- Healthcare
- Energy
- FMCG
- Automobile
- Others

Screen 7 — Capital Allocation

Analyzes:

- Free cash flow
- Capex
- Dividend payout
- Cash generation quality

Screen 8 — Reports

Provides downloadable reports:

- Screener output
- Company reports
- Analytics reports

7. PDF Tearsheet Generation

Purpose

A company tearsheet provides a complete analyst report.

Steps

1. Open Company Profile page
2. Search company ticker

Example:

TCS

3. Select Generate Tearsheet
4. Download PDF

Tearsheet Contains

- Company overview
- Financial summary
- KPI analysis
- Growth trends
- Peer comparison
- Investment indicators

8. API Usage Guide

API Server Start

Command:

```
uvicorn src.api.main:app --port 8000
```

API documentation:

<http://localhost:8000/docs>

API Base URL

<http://localhost:8000/api/v1>

Health Check

Request:

```
curl http://localhost:8000/api/v1/health
```

Response:

```
{  
  "status": "ok",  
  "version": "1.0.0"  
}
```

Get Companies

Request:

```
curl http://localhost:8000/api/v1/companies
```

Returns:

- Company list

- Sector
 - Financial KPIs
-

Company Profile

Example:

```
curl http://localhost:8000/api/v1/companies/TCS
```

Screener API

Example:

```
curl "http://localhost:8000/api/v1/screener?min_roe=15"
```

Returns companies where:

ROE $\geq$ 15%

Sector API

Example:

```
curl http://localhost:8000/api/v1/sectors
```

9. Analytics Engine

Clustering Analysis

The platform groups companies into five financial profiles.

Example clusters:

1. High Quality Compounders

Characteristics:

- High ROE
- Strong growth
- Positive cash flow

2. Defensive Dividend Payers

Characteristics:

- Stable earnings
- High dividend payout

3. Value Cyclical

Characteristics:

- Moderate valuation
- Cyclical growth

4. Distressed / Turnaround

Characteristics:

- Weak historical performance

- Improvement potential

5. Emerging Growth

Characteristics:

- High revenue growth
 - Expansion stage
-

10. Data Quality Framework

The platform includes 14 automated data quality rules.

Checks include:

- Duplicate records
- Missing values
- Invalid financial values
- Foreign key validation
- Data consistency

All rules are tested automatically.

11. Testing Framework

Run complete tests:

```
pytest tests/ -v
```

Expected:

60+ tests passed

0 failures

HTML report:

reports/pytest_report.html

12. Troubleshooting Guide

Issue 1: API Does Not Start

Error:

ModuleNotFoundError: src

Solution:

Run command from project root:

```
N100_Financial_Intelligence_Platform
```

Issue 2: Database Table Missing

Error:

no such table

Solution:

Run ETL:

```
python src/etl/loader.py
```

Issue 3: Streamlit Page Not Loading

Solution:

Restart:

```
streamlit run src/dashboard/app.py
```

Issue 4: Tests Failing

Run:

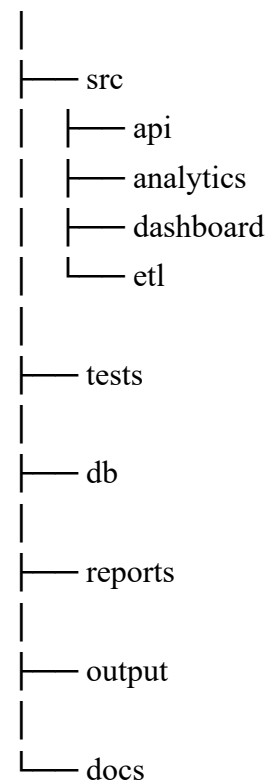
```
pytest tests/ -v
```

Check failed module and logs.

13. Developer Reference

Project Structure

N100_Financial_Intelligence_Platform



14. Final Deliverables

The project includes:

File	Description
cluster_labels.csv	Company clustering output
elbow_plot.png	KMeans validation
correlation_heatmap.png	KPI correlation
outlier_report.csv	Statistical outliers
portfolio_stats.csv	Portfolio statistics
openapi.json	API documentation
pytest_report.html	Test execution report
analyst_guide.pdf	User documentation

15. Conclusion

The N100 Financial Intelligence Platform provides a complete investment analytics ecosystem.

The platform enables analysts to:

- Evaluate companies
- Compare peers
- Identify opportunities
- Analyze financial quality
- Generate reports
- Access insights through APIs

Through automated ETL, analytics, testing, and visualization, the platform delivers reliable and scalable financial intelligence.

End of Analyst Guide

Version 1.0